PART 15 OF 18

Cloud Implementation Comparison

AWS Bedrock AgentCore, Azure AI Foundry, Google Vertex AI, Framework Comparison (LangGraph, CrewAI, AutoGen), Open-Source Stack

ENTERPRISE AI CONTROL ARCHITECTURE

Implementation Guide for Production AI Systems • 2026

15.1 Cloud Platform Comparison for Enterprise AI Agents

The three major cloud providers have each built enterprise-grade AI agent platforms with different architectural philosophies, security models, and maturity levels. The choice of platform has significant implications for security architecture, operational complexity, and vendor lock-in. This analysis provides an objective comparison based on publicly available documentation and reported enterprise deployments as of mid-2026.

15.2 AWS: Amazon Bedrock AgentCore

15.2.1 Platform Overview

AWS Bedrock AgentCore (launched 2025) provides a production-grade agent execution platform deeply integrated with the AWS security ecosystem. AgentCore focuses on enterprise security primitives, making it the most security-complete platform for enterprises already invested in AWS.

15.2.2 Security Capabilities

Capability	Implementation
Identity	AWS IAM with STS AssumeRole; task-scoped session policies; IAM Roles Anywhere for hybrid deployments
Authorization	Amazon Verified Permissions using Cedar; fine-grained, goal-aware policy evaluation
Tool Security	Bedrock AgentCore MCP server management; tool invocation via Lambda (isolated execution)
Memory	AgentCore Memory with DynamoDB backend; encryption at rest with KMS; access logging with CloudTrail
Observability	CloudWatch + CloudTrail + Bedrock-specific traces; integrates with Arize Phoenix via Lambda
Threat Detection	Amazon GuardDuty for ML-based anomaly detection; Macie for sensitive data detection in agent outputs
Network	VPC isolation; PrivateLink for private API access; WAF for agent endpoint protection
Compliance	SOC 2, ISO 27001, HIPAA BAA, FedRAMP; GDPR data residency via region selection

15.2.3 AWS Strengths

- Most mature policy-as-code framework (Cedar) in the industry
- Deep integration with AWS security services provides comprehensive defense-in-depth
- GuardDuty and Macie integration provides automated threat detection
- Best-in-class secret management with Secrets Manager and KMS
- IAM fine-grained permissions provide strong least-privilege implementation
- AgentCore MCP management provides enterprise-grade tool governance

15.2.4 AWS Weaknesses

- Significant vendor lock-in: Cedar, AgentCore APIs, and Bedrock models are AWS-proprietary
- Multi-cloud deployments require significant additional architecture work
- Cost complexity: multiple overlapping services create difficult cost modeling
- GuardDuty anomaly detection models not trained on AI agent behavior patterns

15.3 Microsoft Azure: Azure AI Foundry

15.3.1 Platform Overview

Azure AI Foundry (formerly Azure AI Studio, rebranded 2024) provides an integrated development and deployment platform for enterprise AI agents. Microsoft's strength lies in Entra ID integration, Microsoft Defender ecosystem, and Office 365 data connectivity for enterprise agents.

Capability	Implementation
Identity	Entra ID Workload Identity Federation; Managed Identities for Azure compute; Entra External ID for cross-tenant
Authorization	Azure RBAC + Entra ID Conditional Access; no Cedar equivalent; custom policy requires Azure Policy or OPA
Tool Security	Azure Functions as tool execution; API Management for tool governance; Key Vault for secrets
Memory	Azure AI Search + Cosmos DB for memory; CMK encryption; private endpoints
Observability	Azure Monitor + Application Insights; Microsoft Security Copilot for AI-assisted SOC functions
Threat Detection	Microsoft Defender for Cloud with AI-specific detections; Sentinel for SIEM integration
Network	Private Link; VNet integration; Azure Front Door for WAF
Compliance	Comprehensive: SOC 2, ISO 27001, HIPAA, FedRAMP High, EU AI Act readiness

15.3.2 Azure Strengths and Weaknesses

Strengths

- Best Microsoft 365 integration for enterprise agents accessing email, calendar, Teams, SharePoint
- Microsoft Defender + Sentinel ecosystem for comprehensive threat detection
- Entra ID is the enterprise identity standard for Microsoft-heavy organizations
- Microsoft Security Copilot provides AI-assisted SOC capabilities
- Strong compliance coverage particularly for European enterprises

Weaknesses

- Authorization policy framework less mature than Cedar; requires more custom development
- AI-specific security controls less specialized than AWS at time of analysis

- Foundry platform more development-oriented than production-operations-oriented

15.4 Google Cloud: Vertex AI Agent Builder

15.4.1 Platform Overview

Google Cloud's agent platform (Vertex AI Agent Builder + Gemini models + Cloud IAM) benefits from Google's deep AI research heritage and the A2A protocol as an open standard for agent interoperability. Google Cloud IAM provides workload identity through Workload Identity Federation.

Capability	Implementation
Identity	Workload Identity Federation; Service Account impersonation; SPIFFE-compatible identity
Authorization	Cloud IAM with condition-based access; no Cedar equivalent; VPC Service Controls for perimeter
Tool Security	Cloud Functions / Cloud Run for tool execution; Artifact Registry for tool container management
Memory	Vertex AI Vector Search + Cloud Bigtable; CMEK; VPC Service Controls
Observability	Cloud Logging + Cloud Trace + Cloud Monitoring; Security Command Center for threat detection
Threat Detection	Security Command Center; Chronicle SIEM; Event Threat Detection
Network	VPC Service Controls; Private Google Access; Cloud Armor WAF
Compliance	SOC 2, ISO 27001, HIPAA, FedRAMP; strong GDPR EU region support

15.5 Agent Framework Comparison

Framework	Model	Multi-Agent	Human-in-Loop	Production Maturity	Security Features
LangGraph	Graph-based stateful agents	Yes (multi-node graphs)	Built-in interrupt points	HIGH - widely deployed	Checkpoint/rollback; state persistence; extensible PEP hooks
CrewAI	Role-based agent crews	Yes (crew coordination)	Limited built-in	MEDIUM-HIGH	Role-based access within crew; basic tool governance
AutoGen	Conversational multi-agent	Yes (conversation framework)	Human proxy agent	MEDIUM	Basic authorization hooks; good for research/prototyping

Framework	Model	Multi-Agent	Human-in-Loop	Production Maturity	Security Features
Semantic Kernel	Plugin-based, enterprise-focused	Partial	Process automation integration	HIGH - enterprise use	Strong Azure integration; policy-based plugin governance
Anthropic Agents SDK	Native Anthropic integration	Yes	Built-in approval gates	HIGH - production ready	Deep constitutional AI integration; safety-first design
LlamaIndex Workflows	Event-driven agentic workflows	Yes (multi-agent)	Event-based interrupts	MEDIUM-HIGH	Good observability; extensible security hooks

15.6 Open-Source Security Stack

For enterprises that require vendor independence, maximum control, or highly regulated environments, an open-source security stack can provide comparable security to cloud platform native offerings. The stack requires more engineering investment but provides maximum flexibility.

Layer	Open Source Component	Alternative to	Maturity
Identity	SPIFFE/SPIRE	Cloud Workload Identity	HIGH - CNCF graduated
Authorization	OPA / Rego or OpenFGA	Cedar / Azure Policy	HIGH - widely deployed
Secrets	HashiCorp Vault (OSS)	AWS Secrets Manager	HIGH - enterprise standard
Observability	OpenTelemetry + Grafana + Jaeger	CloudWatch / Azure Monitor	HIGH - CNCF graduated
Security Monitoring	Falco (eBPF) + Elastic SIEM	GuardDuty / Sentinel	HIGH - widely deployed
Agent Orchestration	LangGraph / Apache Airflow	Bedrock AgentCore	HIGH
Vector Memory	Weaviate / Milvus / Qdrant	Vertex Vector Search	HIGH - production ready
AI Observability	Langfuse + Arize Phoenix	CloudWatch Bedrock logs	MEDIUM-HIGH
Policy Engine	OPA Gatekeeper	Verified Permissions	HIGH

Layer	Open Source Component	Alternative to	Maturity
Container Security	Falco + Trivy + Cosign	GuardDuty Container	HIGH